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<td>UNIVERSIDADE DE COIMBRA
FACULDADE DE CIÊNCIAS E TECNOLOGIA
Departamento de Engenharia Informática</td>
<td>Assignment #2
Integração de Sistemas/
Enterprise Application Integration

2025/26
MEI, MSI

Deadline: 2026-04-17</td>
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<td colspan="2"><u>Notas</u>:
A fraude denota uma grave falta de ética e constitui um comportamento não admissível num estudante do ensino superior e futuro profissional. Qualquer tentativa de fraude pode levar à reprovação na disciplina tanto do facilitador como do prevaricador.</td>
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The Model Context Protocol (MCP)

Objectives

- Understand how MCP exposes capabilities to LLM-based agents
- Design Application Programming Interfaces (APIs) that can be consumed by Artificial Intelligence (AI) systems
- Distinguish between actions (tools) and context (resources)
- Create reusable prompt templates
- Build a simple client interface for interacting with an MCP server

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Overview

Large language model (LLM) applications increasingly rely on structured interfaces to interact with external systems. The **Model Context Protocol (MCP)** provides a standard mechanism for exposing capabilities such as tools, resources, and prompts to AI assistants. In this assignment, students will design and implement a small service whose functionality is exposed through MCP and Representational State Transfer (REST). The system will provide a simple domain Application Programming Interface (API) and allow an AI client to interact with it via MCP primitives.

Resources

- What is the Model Context Protocol (MCP)? - Model Context Protocol: https://modelcontextprotocol.io/docs/getting-started/intro
- Model Context Protocol (MCP) Assets | by Abhishek Jaiswal | Medium: https://abhishek-iiit.medium.com/model-context-protocol-mcp-assets-8cae94599875
- Welcome to FastMCP – FastMCP: https://gofastmcp.com/getting-started/welcome
- FastAPI: https://fastapi.tiangolo.com/
- SQLModel: https://sqlmodel.tiangolo.com/
- Java MCP Server - Model Context Protocol: https://modelcontextprotocol.io/sdk/java/mcp-server
- The official Java SDK for Model Context Protocol servers and clients: https://github.com/modelcontextprotocol/java-sdk
- SpringBoot: https://spring.io/projects/spring-boot
- LangChain: Observe, Evaluate, and Deploy Reliable AI Agents: https://www.langchain.com/

MCP Training Part (doesn't count for evaluation)

Before they implement the final solution, students should write a small server to experiment the technologies. They can pick a domain they like for the server, e.g., a pet clinic, a library, a car repair shop, or other. Before starting implementation, students are encouraged to do some preparatory exercises.

1. Create a REST service that has a greet or hello world message.

2. Create an Entity, connect it to a database, such as SQLite or other and add CRUD operations for that entity in the REST service.

3. Expose the same operations on that Entity with MCP tools. Demonstrate that these tools are working properly by connecting an agent with the corresponding MCP client configured. Students may use VS Code as the agent. Students should ensure that core functions are shared both by the REST and the MCP interfaces, as in Figure 1.

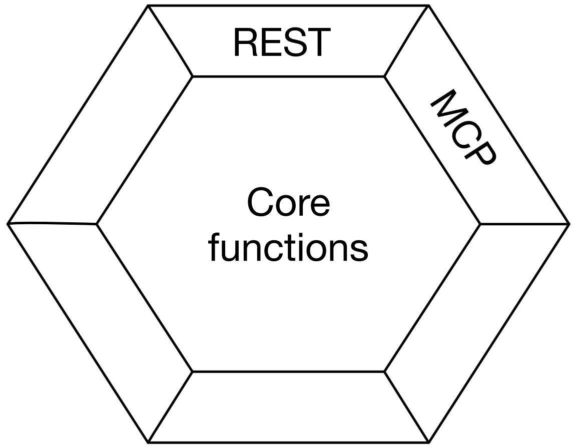

Figure 1 - Server core functions exposed as REST and MCP.

4. Expose a text or pdf file as resource, such as the database model or some internal report.

5. Add a prompt to the MCP server. For example, in a pet shop scenario, the prompt could describe in detail a process for extracting and organizing data about pet consultations.

Description of the Assignment (for Evaluation)

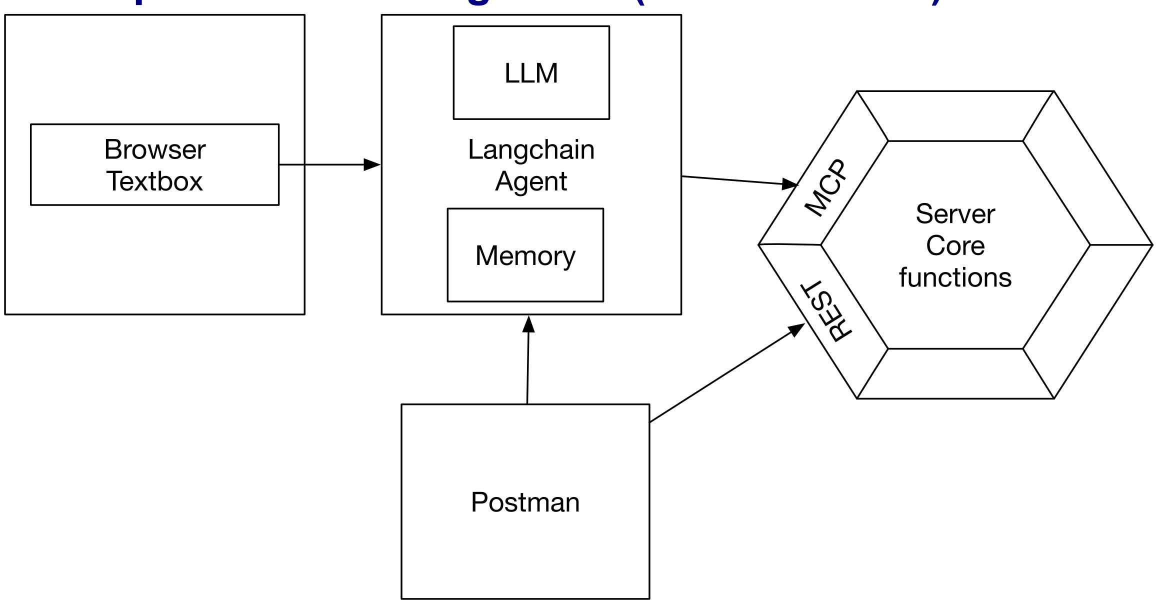

Figure 2 – Overall architecture to build.

The goal of this assignment is to create a web page with a simple text box that allows users to interact with an agent that can resort to an MCP server, as in Figure 2.

Students must select a domain (possibly the same as in the practicing part) with at least two entities related to each other, such as pets and doctors, and expose Create-Read-Update-Delete (CRUD) functionality on these entities via REST and MCP, to enable a simple use of the functionality by AI agents. Students must expose MCP **tools, resources** (at least two), and at least one **prompt** for the agent. The prompt may be used at startup to initialize the LLM. The MCP server should also:

- Expose at least one operation or parameter in one of the operations that raises some exception and that cannot be concluded.
- Expose at least one operation or parameter in one of the operations that returns an error.
- Have two operations similar in nature and output (e.g. add) but where the input parameters and output parameters are slight variations of each other (one accepts and outputs as a strings, the other as a float, for example).

Students may select Java or Python for their core services, REST and MCP implementations. The agent should be written using the Langchain platform. The domain, the tools, and other options should be discussed with the professors. A working example that students can download and improve is provided.

Students should prepare a pdf presentation for 5 minutes, which they will discuss with the professors.

Good work!

Final Delivery

- The work should be done in groups of **2 students**. Both group members must be fully aware of all the details as their defense may occur separately. Work together!
- Grades will be based on the quality of the presentation and defense.
- Students should submit the presentation pdf at Inforestudante. Students must also submit the source code for discussion.
